

JAYDIP VASOYA

Senior Full Stack Engineer | Python, Django, React, AWS | AI Agents and FinTech

jpvasoya444@gmail.com | +91 7487029363 | linkedin.com/in/jaydip-vasoya | github.com/jp-vasoya-02 | Ahmedabad, India

PROFESSIONAL SUMMARY

Senior Full Stack Engineer with over 6 years of experience building FinTech platforms and production AI products. Recent work includes a Large Language Model (LLM) research agent on Claude and AWS Bedrock, a desktop AI assistant that reads and controls live trading charts through a Model Context Protocol (MCP) server, a real time voice AI assistant, and a market data pipeline that processes the US options (OPRA) firehose. Core stack: Python, Django, React, TypeScript, and AWS. Comfortable owning a product from architecture to production and working directly with clients.

TECHNICAL SKILLS

Programming Languages: Python, TypeScript, JavaScript, SQL

Backend: Django, Django REST Framework, Django Channels, FastAPI, Flask, Node.js, Celery, WebSockets, REST APIs

AI and LLM Engineering: Anthropic Claude, AWS Bedrock, Google Gemini (Live API), OpenAI, LangGraph, LangChain, DeepAgents, LangFuse, MCP (Model Context Protocol), AI Agents, Tool Calling, Multi Agent Systems, RAG (Retrieval Augmented Generation), Prompt Caching, Prompt Engineering, Voice AI (ElevenLabs), Hugging Face, Stable Diffusion, LoRA

FinTech: Real time market data, OPRA options feed, options flow and dark pool analytics, WebSocket data feeds, TradingView integration, trading platforms, Stripe subscription billing

Frontend and Desktop: React.js, Next.js, Electron, React Native, Redux, Zustand, Tailwind CSS, Material UI, Three.js

Databases: PostgreSQL, Redis (Streams, RediSearch), MySQL, DynamoDB, Amazon RDS

Cloud and DevOps: AWS (EC2, RDS, S3, SES, Bedrock, Lambda, ECS, ECR, API Gateway, AppSync, SNS, CloudFormation, CloudFront, CodePipeline), Docker, Docker Compose, Kubernetes, Nginx, GitHub Actions, CI/CD, Microservices, PM2, Grafana

Testing and Quality: Pytest, unit and integration testing, code review

Soft Skills: Technical leadership, team mentoring, client communication, problem solving

PROFESSIONAL EXPERIENCE

Senior Full Stack FinTech and AI Engineer

January 2025 to Present

Freelance, Remote

- Serve as the long term engineering partner for an options analytics company, delivering **four connected products**: a web analytics platform, a real time market data pipeline, a desktop AI app, and a voice AI assistant.
- Built the platform's AI research agent with **LangGraph and DeepAgents on Claude through AWS Bedrock**, with token streaming over WebSockets, 16 tools, Postgres backed conversation state, and prompt caching.
- Engineered a pipeline that ingests a vendor's **OPRA options firehose** and dark pool trades into **Redis Streams**, using bounded queues, batched writes of 5,000 messages, and 60 second aggregation across 8 parallel jobs.
- Shipped a cross platform Electron desktop app in which an AI agent reads and controls live TradingView charts through an **MCP server exposing 105 tools**.
- Developed a real time voice AI assistant on Django Channels with Gemini Live and ElevenLabs and metered usage billing, covered by **more than 1,500 automated tests**.
- Hardened production infrastructure, including memory limits and an eviction policy for a Redis instance holding **over 10 million keys**, plus Nginx and Unicorn tuning.

Senior Software Engineer and Partner

January 2020 to January 2025

LNx Cloud Technology

- Led the architecture and development of scalable cloud applications for clients in FinTech, SaaS, and social media.
- Built real time systems with WebSockets and Django Channels that powered live market data feeds and chat products serving **thousands of concurrent users**.
- Improved backend performance and integrated AWS services to achieve high availability, security, and monitoring across production systems.
- Mentored junior developers, ran code reviews, and set engineering standards that improved delivery speed and code quality across the team.

KEY PROJECTS

AI Options Analytics Platform

Technologies: Python, Django, Django REST Framework, Django Channels, Celery, PostgreSQL, Redis, AWS (EC2, RDS, S3, SES, Bedrock), LangGraph, LangFuse, Stripe

- Core engineer on a subscription platform with four pricing tiers and **over 130 analytics pages** for options flow, dark pool, and volatility data, updated live over more than 100 WebSocket routes.
- Took the AI research agent through **eight versions**, moving from semantic routing to a multi agent design and then to a single deep agent on Claude, traced end to end with LangFuse.
- Added agent guardrails: server side limits on tool arguments, prompt leak detection, automatic context summarization, and a deterministic scoring engine so the model never calculates figures itself.
- Built **per request LLM cost metering** with price snapshots, a page level AI assistant that analyzes the charts a user is viewing, and Stripe subscription billing.

Real Time Options Flow and Dark Pool Pipeline

Technologies: Python, Redis Streams, RedisSearch, WebSockets, Multithreading, AWS S3, PM2

- Designed a stream writer that holds up under traffic bursts, with a **bounded queue of 50,000 items** and a connection pool per stream, pipelined batch writes, and alerts on dropped messages.
- Built the dark pool pipeline with two tier trade sentiment (bid and ask spread first, tick rule as fallback), a correction pass five minutes after each trade, and atomic alert cooldowns.
- Added burst detection on 200 ms buckets and self healing that restarts a stalled feed within three minutes.

AI Desktop Trading Assistant

Technologies: Electron, React, TypeScript, Tailwind CSS, Anthropic SDK, AWS Bedrock, Gemini, MCP, Chrome DevTools Protocol

- Built the React and TypeScript interface and the agent loop, with **native tool calling** for Anthropic, AWS Bedrock, and Gemini models, token streaming, prompt caching, and request cancellation.
- Integrated an MCP server that exposes **105 tools in 17 groups** over the Chrome DevTools Protocol, letting the assistant read indicators, switch symbols, draw levels, and create alerts on a live chart.
- Enforced user approval for destructive actions at a single gateway, and shipped signed, notarized builds for macOS and Windows with in app auto update.

Voice AI Trading Assistant

Technologies: Python, Django Channels, Celery, PostgreSQL, Redis, Gemini Live API, ElevenLabs, Stripe, Docker, Nginx, Pytest

- Built a browser based voice assistant with a live transcript on Django Channels, with **Gemini Live** native audio and **ElevenLabs** engines behind one interface so the WebSocket layer does not depend on the provider.
- Exposed **56 tools** to the model and ran the tool calls of a turn concurrently with asyncio to keep spoken replies fast.
- Implemented usage metering with weekly and monthly spend caps and prepaid credits, and a Docker Compose deployment behind Nginx with rate limiting, JWT authentication, and MFA.

Additional Projects

- **Marketing and SEO website:** Next.js 16 and React 19 static export with lazy loaded React Three Fiber 3D scenes, JSON-LD structured data, and build checks for bundle size, SEO rules, and WCAG AA contrast.
- **Real time stock market application:** React, TypeScript, and Django platform that consumes a global exchange WebSocket feed and streams updates to thousands of concurrent clients, with role based access control.
- **AI image generation system:** FastAPI service running fine tuned LoRA models on Stable Diffusion, with scalable GPU inference infrastructure on AWS for high resolution output.
- **Social media platform for mobile and web:** React Native and progressive web apps for media sharing, chat, shopping, and community Q&A; led the migration to a serverless AWS backend (AppSync, Lambda, DynamoDB).
- **SDLC management platform:** NX monorepo frontend in React and Material UI with Django REST Framework APIs on AWS.
- **Encrypted real time chat application:** Django Channels chat with presence, read receipts, and RSA end to end encryption.

EDUCATION

Bachelor of Computer Application, Computer Science

Saurashtra University, Rajkot | CGPA: 8.75 with First Distinction

October 2019 to March 2022

Bachelor's Degree, Information Technology

LDRP Institute of Technology and Research, Gandhinagar | Kadi Sarva Vishwavidyalaya (Kadi University)

July 2017 to August 2019